

Electrical Wire Types for Electrical Systems

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Electrical Wire Types for Electrical Systems

This presentation explores common electrical wire types, their properties, and application.

Vulcanized Indian Rubber (V.I.R.) Wires

Construction & Features

Rubber insulation

Tinned conductor

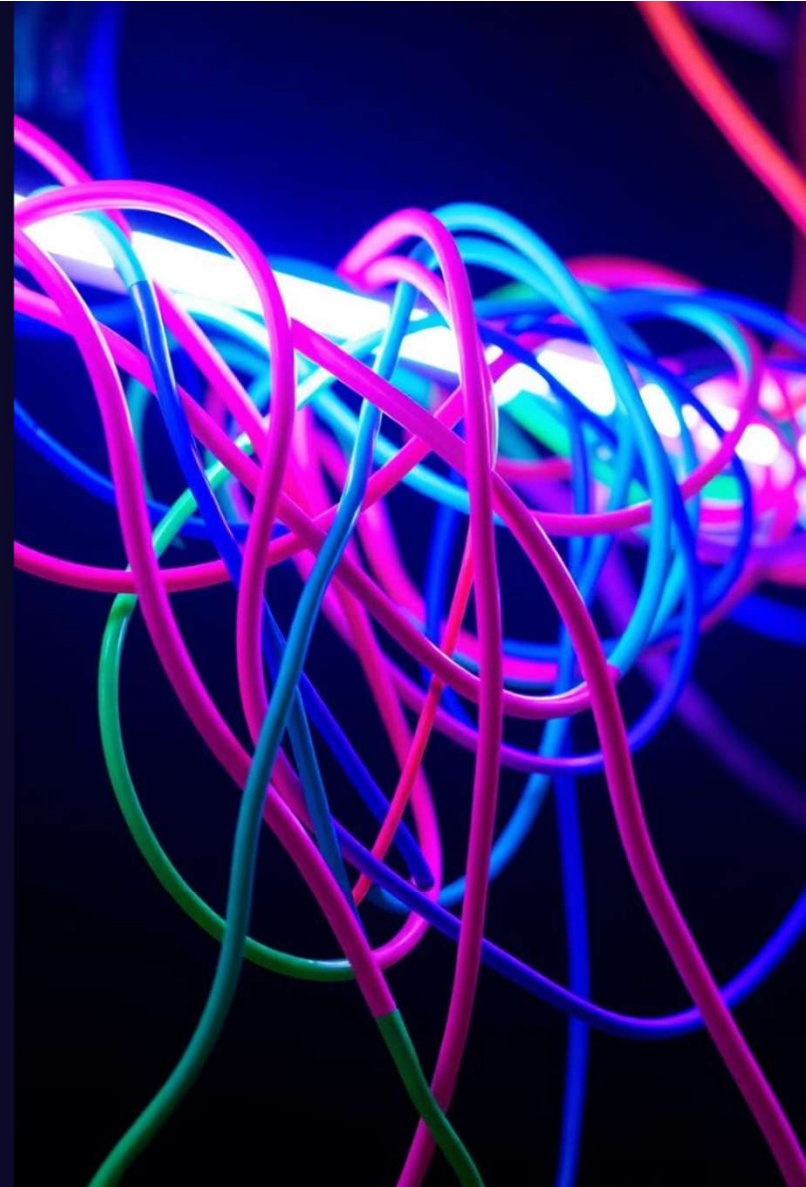
Cotton braiding

Wax coating

Applications

Low/medium
voltage

Older installations



Tough Rubber Sheathed (T.R.S.) Wires

Construction & Features

V.I.R. with tough rubber sheath

Moisture, wear, and damage resistance

Single, twin, and three-core versions

Applications

Industrial and commercial installations

Temporary outdoor wiring

Poly Vinyl Chloride (P.V.C.) Wires

1

Most Common

PVC thermoplastic insulation

Moisture, chemical, and wear resistance

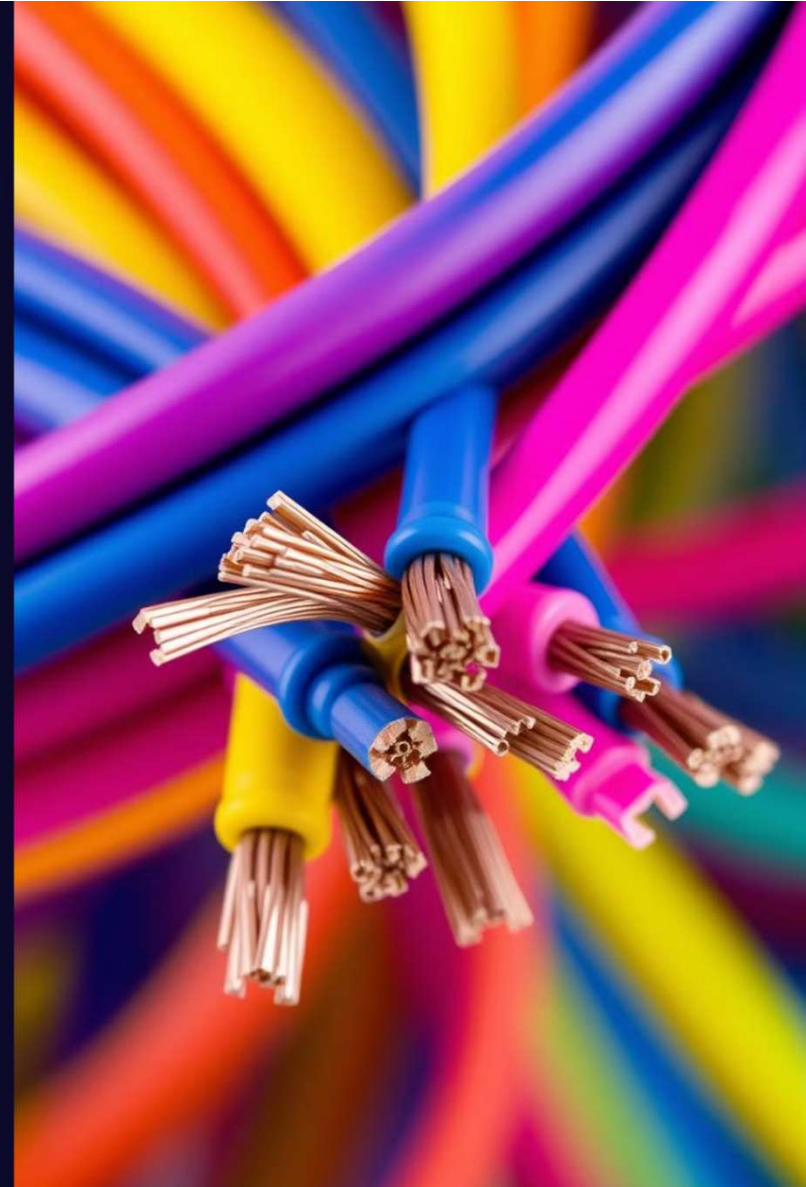
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Widely Used

Household wiring

Industrial electrical systems

Indoor and outdoor applications



Lead Alloy Sheathed Wires

Construction & Features

Outer lead sheath for protection

Moisture-prone environments

0.12 cm thick sheath

Applications

Underground installations

Industries and damp environments

Weatherproof Wires

Construction & Features

Fibrous yarn saturated with waterproof compound

Rain, humidity, and UV radiation resistance

Applications

Service connections from power lines to buildings

Agricultural applications and outdoor lighting

Flexible Wires

Construction & Features

Multiple fine strands

PVC insulation

Frequent movement applications

Applications

Household appliances

Automotive wiring

Portable electrical devices

Choosing the Right Wire

1

Insulation Material

Determines resistance to moisture, heat, and chemicals

2

Mechanical Strength

Impact resistance, flexibility, and durability

3

Environmental Conditions

Exposure to moisture, temperature, and UV radiation

